

3.6 Momentum

Further Mechanics · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Impulse is $\int F dt = \text{change in momentum}$. Momentum is conserved for an isolated system; kinetic energy is conserved only in elastic collisions. The coefficient of restitution e compares relative separation speed to approach speed along the line of impact, ranging from 0 to 1 in usual collision models.

Worked example: follow the reasoning

If two equal masses exchange velocities in a one-dimensional elastic collision, both momentum and kinetic energy are conserved.

Structured practice

Answer in the space provided. Show working and units where relevant.

1 A 1 kg trolley at 4 m s^{-1} hits a stationary 1 kg trolley; they stick. Find common velocity and kinetic-energy change. [3]

Final answer: _____

Answers and explanation

Question 1 [3 marks]

Momentum gives $(1)(4)=(2)v$, so $v=2 \text{ m s}^{-1}$ [2]; initial KE 8 J, final 4 J, change -4 J [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Further Mathematics 9231, 2026–2027.

<https://www.cambridgeinternational.org/Images/697357-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.